

This report documents a local network security assessment conducted as part of the Oasis Infobyte Cyber Security Internship. The objective of the engagement was to evaluate the exposed network surface of a Windows 11 workstation using Nmap, a widely used open-source network scanning utility, and to translate the resulting technical data into actionable security findings.

Network security assessments of this kind are a standard first step in any broader security evaluation. Before an organization can defend a system, it must first understand what is reachable on that system — which ports are open, which services are listening, and what software versions are in use. Unpatched or unnecessarily exposed services are a leading cause of compromise across enterprise and personal computing environments alike, which makes this type of reconnaissance-driven analysis directly relevant to real-world security operations.

The assessment target was the local host itself (127.0.0.1), scanned entirely from within the same machine with no traffic directed at any external third-party system. Three scan phases were performed: a default TCP port scan, a service and version detection scan (-sV), and an operating system fingerprinting scan (-sO). All testing was performed with explicit authorization over the tester's own device, and no exploitation, data extraction, or service disruption was attempted at any stage.

Local Network Security Assessment Using Nmap

An Authorized Host-Level Security Evaluation and Reconnaissance Report

The scan identified four open TCP ports: 135 (Microsoft-RPC), 445 (SMB / Microsoft-DS), 3306 (MySQL), and 6646 (unidentified service). Nmap's OS fingerprinting engine correctly identified the host as running Microsoft Windows 11 (24H2–25H2). Each open port was subsequently analyzed for its associated security posture, and the findings are documented in detail in Sections 8 through 11 of this report, along with corresponding recommendations.

Overall, the assessment demonstrates a practical, end-to-end security evaluation workflow — from reconnaissance and enumeration through risk classification and remediation guidance — and reflects the type of host-level analysis performed during the early stages of a professional security assessment engagement.

PREPARED BY

Akshita Kaul

ACADEMIC PROGRAM

B.Tech, Computer Science and Engineering (Cybersecurity)

INSTITUTION

Amity University, Mumbai

INTERNSHIP / ORGANIZATION

Oasis Infobyte Cyber Security Internship

ASSESSMENT SCOPE

Localhost (127.0.0.1) — Authorized Internal Testing

REPORT DATE

July 7, 2026

Introduction to Cybersecurity Assessment

What Vulnerability Assessment Means

A vulnerability assessment is a systematic process of identifying, classifying, and prioritizing weaknesses in a computer system, network, or application. Unlike penetration testing, which actively attempts to exploit a weakness to demonstrate impact, a vulnerability assessment focuses on discovery and analysis: mapping out what exists on a system and evaluating whether that configuration introduces unnecessary risk.

Why Organizations Perform Network Scanning

Organizations scan their own networks for several reasons: to maintain an accurate inventory of exposed services, to verify that firewall and hardening policies are being enforced correctly, to detect configuration drift after software updates or changes, and to satisfy compliance and audit requirements. Regular scanning transforms security from a reactive activity into a proactive, measurable process.

The Role of Nmap in Cybersecurity

Nmap (Network Mapper) is one of the most widely adopted open-source tools for network discovery and security auditing. It supports host discovery, port scanning, service and version detection, and operating system fingerprinting. Security professionals rely on Nmap during the reconnaissance phase of an assessment because it provides a fast, reliable, and well-documented method for building an accurate picture of a target's network surface before any deeper analysis takes place.

Scanning Versus Attacking

It is important to distinguish between scanning and attacking. Scanning is a passive-to-active reconnaissance activity that queries a system to determine which services are available and how they respond — it does not attempt to alter, damage, or gain unauthorized access to a system. An attack, by contrast, involves actively exploiting an identified weakness to achieve unauthorized access, disruption, or data exposure. This assessment involved scanning and analysis only; no exploitation techniques were used at any point.

Project Objectives

This assessment was structured around six core objectives, each building on the previous to move from raw technical data toward actionable security guidance:

- **Network Reconnaissance** — Establish an accurate baseline of the host's exposed network surface.
- **Port Discovery** — Identify all open TCP ports accessible on the target system.
- **Service Identification** — Determine which services and software versions are bound to each open port.
- **OS Detection** — Fingerprint the underlying operating system and build/version details.
- **Risk Analysis** — Evaluate each discovered service against known risk factors and exposure scenarios.
- **Security Recommendations** — Provide practical, prioritized remediation guidance for each finding.

Tools and Technologies

Nmap 7.99

Nmap is an open-source network scanning engine used to perform host discovery, port scanning, service/version detection, and OS fingerprinting. Version 7.99 was used for all scans in this assessment, executed from the Windows Command Prompt.

Windows 11 Environment

The target and scanning host were the same machine, running Windows 11 (build 26200, 24H2–25H2 branch). Testing entirely within a single Windows 11 environment allowed the assessment to focus on host-level exposure without introducing external network variables.

Command Prompt

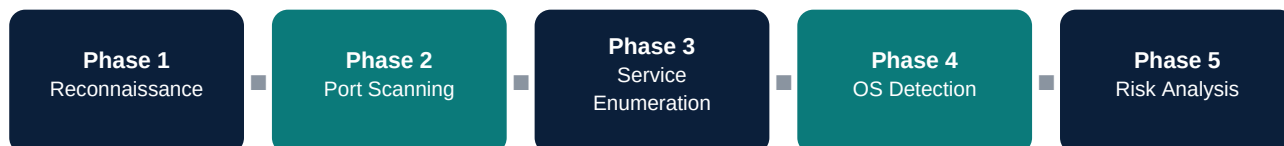
All Nmap commands were executed through the native Windows Command Prompt (cmd.exe), which provided direct console output for each scan phase.

Core Networking Concepts Applied

- **IP Address** — A numerical identifier assigned to a device on a network; this assessment used the loopback address 127.0.0.1.
- **Ports** — Logical endpoints (0–65535) through which a device sends and receives network traffic for specific services.
- **Services** — Software processes bound to a port that respond to incoming requests, such as file sharing or database access.
- **TCP (Transmission Control Protocol)** — A connection-oriented protocol that establishes a reliable session before data is exchanged; the default scan type used throughout this assessment.

Methodology

The assessment followed a structured five-phase methodology, moving from initial reconnaissance through to final risk analysis. This staged approach mirrors the workflow used in professional network security assessments and ensures that no phase is skipped or performed out of sequence.



Phase 1 — Reconnaissance

The target was defined as the local host (localhost / 127.0.0.1) and the scanning environment (Windows 11, Command Prompt, Nmap 7.99) was confirmed and prepared.

Phase 2 — Port Scanning

A default Nmap scan (`nmap localhost`) was executed to enumerate all open TCP ports on the target and establish the initial attack surface.

Phase 3 — Service Enumeration

A version detection scan (`nmap -sV localhost`) was performed against each open port to identify the specific service and, where possible, its software version.

Phase 4 — Operating System Detection

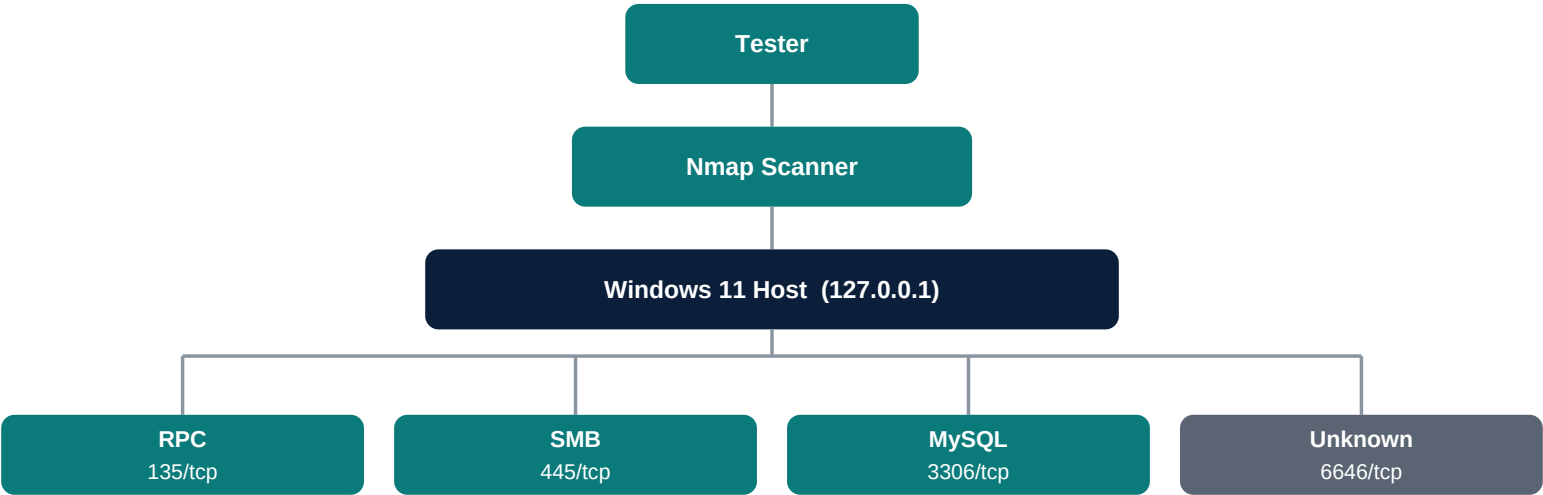
An OS fingerprinting scan (`nmap -O localhost`) was run to determine the underlying operating system, providing context for how each identified service is typically configured and secured on that platform.

Phase 5 — Risk Analysis

Each open port and its associated service were evaluated individually against known exposure patterns and industry-recognized risk factors, producing the findings documented in Sections 8 through 11.

Scan Topology

The diagram below summarizes the assessment path: the tester invoked Nmap against the local host, which in turn exposed four services for analysis.



Scan Results and Evidence

The following console output was captured directly from the assessment host during each scan phase. Output has been reproduced as-recorded, with only the command prompt path retained for context.

1. Basic Port Scan

```
C:\Users\VICTUS>nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-07 14:06 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00047s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
6646/tcp   open  unknown

Nmap done: 1 IP address (1 host up) scanned in 6.05 seconds
```

Figure 1. Default Nmap TCP port scan of localhost, showing four open ports.

The default scan identified four open TCP ports. No filtered or unusual states were reported; all remaining ports outside the shown range returned a closed/reset state, indicating a relatively small exposed surface for this host.

2. Service Version Detection

```
C:\Users\VICTUS>nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-07 14:08 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00018s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
445/tcp    open  microsoft-ds?
3306/tcp   open  mysql        MySQL 8.0.39
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.35 seconds
```

Figure 2. Service and version detection scan, confirming service banners for ports 135, 445, and 3306.

Version detection confirmed Microsoft RPC on port 135, an unconfirmed SMB/Microsoft-DS banner on port 445, and MySQL 8.0.39 on port 3306. Nmap's service probes did not return a positive banner match for every port in this scan window, which is expected behavior when a service does not respond to the specific probes Nmap sends.

3. Operating System Detection

```
C:\Users\VICTUS>nmap -O localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-07 14:11 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00046s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
3306/tcp   open  mysql
Device type: general purpose
Running: Microsoft Windows 11
OS CPE: cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 11 24H2 - 25H2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.06 seconds
```

Figure 3. OS fingerprinting scan, correctly identifying the host as Microsoft Windows 11 (24H2–25H2).

The OS detection engine classified the host as a general-purpose device running Microsoft Windows 11, build range 24H2–25H2, with a network distance of zero hops — consistent with a loopback (localhost) scan.

Detailed Security Analysis

Each open port identified during the assessment was individually analyzed for its function, associated security considerations, and recommended hardening actions. Severity ratings reflect the service's general risk profile in a typical exposure scenario and are not a claim that the specific host was found to be actively compromised or misconfigured.

Finding NS-001 Port 135/tcp — Microsoft RPC (msrpc)		MEDIUM
Description	Port 135 hosts the Microsoft Remote Procedure Call (RPC) endpoint mapper, a core Windows component that allows local and remote processes to locate and communicate with RPC-based services, including DCOM and various system management interfaces.	
Security Risk	RPC has historically been associated with several critical remote code execution vulnerabilities (such as those exploited by the Blaster and Conficker worms). Even where no active vulnerability exists, an exposed RPC endpoint mapper can reveal information about other services running on the host to an attacker performing reconnaissance.	
Potential Impact	If exposed beyond the local machine and left unpatched, this service could be leveraged for information disclosure or, in the case of an unpatched vulnerability, remote code execution.	
Recommendation	Ensure Windows is fully patched, restrict RPC exposure to trusted networks only via host firewall rules, and disable DCOM/RPC-dependent features that are not actively required.	

Finding NS-002 Port 445/tcp — SMB / Microsoft-DS		HIGH
Description	Port 445 hosts the Server Message Block (SMB) protocol, used by Windows for file sharing, printer sharing, and inter-process communication over a network.	
Security Risk	SMB has been the vector for some of the most damaging vulnerabilities in recent history, most notably EternalBlue (MS17-010), which enabled the WannaCry and NotPetya ransomware outbreaks. SMB is also a common target for lateral movement once an attacker has a foothold on a network.	
Potential Impact	An unpatched or unnecessarily exposed SMB service can allow ransomware propagation, unauthorized file access, or full remote code execution depending on the specific vulnerability present.	
Recommendation	Apply all available SMB-related security patches, disable SMBv1 if still enabled, restrict SMB to trusted internal networks using firewall rules, and disable the service entirely on hosts that do not require file or printer sharing.	

Detailed Security Analysis (continued)

Finding NS-003 | Port 3306/tcp — MySQL Database (MySQL 8.0.39)

MEDIUM

Description	Port 3306 is the default listening port for a MySQL database server, identified here as version 8.0.39. Databases on this port typically store structured application or development data.
Security Risk	Database services are high-value targets because they often contain sensitive or business-critical data. Risks include weak or default credentials, outdated versions with known CVEs, and exposure of the service beyond the intended local development scope.
Potential Impact	Unauthorized access to a MySQL instance could result in data theft, data manipulation, or use of the database server as a pivot point for further access within the environment.
Recommendation	Confirm the database is bound only to localhost unless external access is explicitly required, enforce strong unique credentials, keep MySQL updated to the latest patched release, and enable connection logging for auditing purposes.

Finding NS-004 | Port 6646/tcp — Unidentified Service

LOW

Description	Port 6646 responded as open during the scan but Nmap was unable to positively identify the service or application bound to it during this assessment window.
Security Risk	Unidentified or non-standard services represent an analysis gap: without confirming what is listening on a port, it is not possible to fully assess its associated risk. Unknown services can sometimes correspond to development tools, background applications, or, in rarer cases, unwanted or malicious software.
Potential Impact	Until identified, this port should be treated as an open question rather than a confirmed low-risk finding; it could range from a benign local application to an unnecessary and unmonitored exposure.
Recommendation	Identify the owning process (for example, using Windows tools such as netstat -ano combined with Task Manager or Resource Monitor), confirm whether the associated application is required, and close or firewall the port if the service is not needed.

Risk Assessment Summary

The table below consolidates all findings identified during this assessment, providing a single-view reference for prioritization and remediation tracking.

Finding ID	Port	Service	Risk Level	Impact	Recommendation
NS-001	135	Microsoft RPC	Medium	Information disclosure; potential RCE if unpatched	Patch OS; restrict RPC exposure
NS-002	445	SMB / Microsoft-DS	High	Ransomware propagation; remote code execution	Patch SMB; disable SMBv1; restrict/disable service
NS-003	3306	MySQL 8.0.39	Medium	Data theft or manipulation; pivot point	Bind to localhost; strong credentials; keep updated
NS-004	6646	Unidentified	Low	Unknown — requires further identification	Identify owning process; close if unnecessary

Security Recommendations

Patch Management

Maintain a consistent patching cadence for the Windows operating system and any installed services, particularly SMB and RPC components, which have a history of critical vulnerabilities. Enable automatic updates where operationally acceptable.

Firewall Configuration

Configure host-based firewall rules to restrict access to identified services to only the networks and hosts that require it. Services that do not need to be reachable from other devices should be bound to the loopback interface or blocked entirely at the firewall.

Removing Unnecessary Services

Disable or uninstall services that are not actively required, such as file sharing (SMB) on machines that do not need it, to reduce the overall attack surface.

Strong Authentication

Enforce strong, unique credentials for all services that accept authentication, particularly the MySQL database instance, and consider multi-factor authentication where supported.

Regular Security Assessments

Perform periodic scans of the host to detect newly opened ports, unexpected services, or configuration drift over time, and treat this assessment as a repeatable baseline rather than a one-time exercise.

Limitations

This assessment was intentionally scoped to provide a clear, controlled learning exercise. The following limitations should be considered when interpreting the findings:

- Only the local machine (127.0.0.1) was tested; no other hosts, subnets, or external-facing assets were assessed.
- No exploitation, credential testing, or intrusive techniques were performed at any stage of this engagement.
- Results reflect the system configuration at the time of scanning and may change following software updates, patches, or configuration changes.
- All testing was conducted with proper authorization on a machine owned and controlled by the assessor.

Conclusion

This assessment successfully demonstrated a complete, structured network security evaluation workflow — from initial reconnaissance and port scanning through service enumeration, operating system detection, and formal risk analysis. Four open TCP ports were identified and individually evaluated. The assessment identified services with different risk levels based on their exposure potential, each paired with a corresponding remediation recommendation.

Beyond the technical output, the exercise reinforced several core cybersecurity skills relevant to a professional security role: interpreting raw scan data, correlating discovered services with known risk patterns, communicating findings in a structured and audience-appropriate format, and translating technical observations into practical recommendations.

Network reconnaissance of this kind forms the foundation of nearly every subsequent security activity, including vulnerability management, penetration testing, and incident response. A clear understanding of what is exposed on a system — and why that exposure matters — is a prerequisite for defending it effectively.

Learning Outcomes

This project provided direct, applied practice with the following core cybersecurity skills:

- **Network Reconnaissance** — Mapping a host's exposed network surface as the starting point of a security assessment.
- **Port Scanning** — Using Nmap to enumerate open TCP ports and interpret scan output accurately.
- **Service Enumeration** — Identifying services and software versions bound to open ports.
- **Risk Identification** — Evaluating discovered services against known exposure patterns and classifying risk levels.
- **Security Documentation** — Structuring technical findings into a clear, audience-appropriate report with actionable recommendations.
- **Nmap Usage** — Practical, hands-on experience with default scans, version detection (-sV), and OS fingerprinting (-O).

Project Repository

The complete project — including raw scan output, supporting screenshots, the security analysis, and this final report — is version-controlled and publicly available for review.

GitHub Repository	https://github.com/yourusername/CyberSecurity-Task1-NetworkScanning
Repository Contents	Nmap scan results • Assessment screenshots • Security analysis notes • Final PDF report

References

- Nmap Official Documentation — <https://nmap.org/book/man.html>
- Nmap Reference Guide — <https://nmap.org/book/toc.html>
- Microsoft Learn: SMB Security Overview — <https://learn.microsoft.com/windows-server/storage/file-server/smb-overview>
- Microsoft Learn: Remote Procedure Call (RPC) — <https://learn.microsoft.com/windows/win32/rpc/rpc-start-page>
- MySQL 8.0 Reference Manual — <https://dev.mysql.com/doc/refman/8.0/en/>
- NIST SP 800-115: Technical Guide to Information Security Testing and Assessment — <https://csrc.nist.gov/publications/detail/sp/800-115/final>